



## R2.2 How fast? The rate of chemical change

### Reactivity 2 · How much, how fast and how far? · IB Diploma · Concept guide and original exam-style practice

#### Understand the concept

Collision theory states that a chemical reaction occurs only when reactant particles collide with sufficient energy (at least the activation energy,  $E_a$ ) and with the correct orientation; reaction rate increases with increasing concentration/pressure (more frequent collisions), increasing temperature (more particles have energy  $\geq E_a$ , shown by the Maxwell-Boltzmann distribution shifting and broadening), increasing surface area of a solid (more particles exposed for collision), and the presence of a catalyst (which provides an alternative pathway with lower  $E_a$ , increasing the fraction of successful collisions without being consumed). The rate expression,  $\text{rate} = k[A]^a[B]^b$ , relates rate to reactant concentrations raised to their respective (experimentally determined) orders of reaction, and can be found from initial rates data by comparing experiments in which only one concentration is varied at a time.

#### Worked example: follow the reasoning

The initial rate of a reaction  $A + B \rightarrow \text{products}$  was measured for different starting concentrations:  
Experiment 1:  $[A] = 0.10 \text{ mol dm}^{-3}$ ,  $[B] = 0.10 \text{ mol dm}^{-3}$ ,  $\text{rate} = 2.0 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$ .  
Experiment 2:  $[A] = 0.20 \text{ mol dm}^{-3}$ ,  $[B] = 0.10 \text{ mol dm}^{-3}$ ,  $\text{rate} = 8.0 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$ .  
Experiment 3:  $[A] = 0.20 \text{ mol dm}^{-3}$ ,  $[B] = 0.20 \text{ mol dm}^{-3}$ ,  $\text{rate} = 8.0 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$ .

Deduce the rate equation.

Step 1: Compare Exp 1  $\rightarrow$  Exp 2:  $[A]$  doubles ( $B$  constant), rate increases by  $\times 4$  ( $2.0 \rightarrow 8.0$ ), so  $\text{rate} \propto [A]^2$ ; order with respect to  $A = 2$ .

Step 2: Compare Exp 2  $\rightarrow$  Exp 3:  $[B]$  doubles ( $A$  constant), rate is unchanged ( $\times 1$ ), so order with respect to  $B = 0$ .

Step 3: Overall order =  $2 + 0 = 2$ .

Answer:  $\text{rate} = k[A]^2$ , second order overall (second order in  $A$ , zero order in  $B$ ).

#### Quick check · multiple choice

Which change would increase the rate of a reaction between magnesium ribbon and dilute hydrochloric acid, without changing the activation energy of the reaction?

- A Using magnesium powder instead of ribbon
- B Adding a suitable catalyst
- C Decreasing the temperature
- D Using a more dilute acid

Write your choice: \_\_\_\_\_

### Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A student investigates the decomposition of hydrogen peroxide catalysed by manganese(IV) oxide:  $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ . [8]
  - (a) Sketch (describe in words) the shape of the Maxwell-Boltzmann distribution curve and explain, with reference to activation energy, why only some collisions between particles result in a successful reaction. [3]
  - (b) Explain, in terms of activation energy, how the manganese(IV) oxide catalyst increases the rate of decomposition of hydrogen peroxide. [2]
  - (c) The experiment is repeated at a higher temperature. Explain, using the Maxwell-Boltzmann distribution, why the rate of reaction increases. [3]

Final answer: \_\_\_\_\_



## Answers and explanation

### Question 1 [8]

(a) The Maxwell-Boltzmann distribution is asymmetric: it rises from the origin (zero particles have zero energy), reaches a peak (most probable energy), and has a long tail extending to high energy; the area under the curve represents the total number of particles. Only particles with energy equal to or greater than the activation energy,  $E_a$  (a value marked towards the higher-energy tail of the curve), can react on collision; particles with energy below  $E_a$  do not have enough energy to break the necessary bonds even if they collide, so most collisions are unsuccessful. [3]

(b) The catalyst provides an alternative reaction pathway/mechanism with a lower activation energy; this means a greater proportion/fraction of collisions now have sufficient energy ( $\geq$  the new, lower  $E_a$ ) to react successfully, increasing the rate without the catalyst being consumed. [2]

(c) At a higher temperature, the Maxwell-Boltzmann distribution shifts to the right and flattens/broadens (average kinetic energy increases); the area under the curve to the right of  $E_a$  (representing the fraction of particles with energy  $\geq E_a$ ) increases significantly, so a greater proportion of collisions are successful, and particles also collide more frequently, together increasing the rate of reaction. [3]

### Quick check answer

A. Using powder increases the surface area of the magnesium exposed to the acid, increasing the frequency of successful collisions, without changing the activation energy (unlike a catalyst, which lowers  $E_a$ ).

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.